

Question

A new compound **X** was prepared by hydrothermal reaction of PbO_2 and Ag_2O in a silver crucible for 7 days. The following information is related to **X**:

(a) 1.000 g of **X** was added to 25.00 mL of 0.1015 mol/L oxalic acid solution acidified, and after sufficient reaction (the existing forms of each element in the product: $\text{Ag} \rightarrow \text{Ag}^+$; $\text{Pb} \rightarrow \text{Pb}^{2+}$) (reaction 1), it was titrated with 0.02030 mol/L acidic KMnO_4 solution (reaction 2), consuming 21.84 mL.

(b) The lead in 0.500 g of **X** was completely converted to PbSO_4 precipitate, and its weight was measured to be 0.286 g.

(c) Structure determination showed that there are no hydroxyl groups or water molecules in **X**, and the ratio of the number of each atom in **X** is a simple integer ratio; conductivity measurement showed that **X** is a superconductor.

Recently, it has been found that **X** can also be obtained by mixing and heating Ag_2O , PbO_2 and a small amount of activated carbon in vacuum (reaction 3).

Let the sum of the stoichiometric numbers of the ionic equations of reactions 1 and 2 be a and b, respectively, and the sum of the stoichiometric numbers of the chemical equation of reaction 3 be c, then $(a+b)/c$ is approximately

A. All other options are incorrect

B. 2.67

C. 3.67

D. 1.33

E. 1.67

F. 2.75

G. 3.45

H. 4.09

I. 4.83

J. 3.33

K. 1.88

L. 2.37

M. 3.42

N. 3.50

O. 2.83

P. 2.17

Q. 1.92

R. 5.78

S. 6.10

Answer

Correct Answer: **C**

Detailed Explanation

The total amount of electrons transferred by oxalic acid is:

$$2 \times 25.00 \text{ mL} \times 0.1015 \text{ mol/L} = 5.075 \times 10^{-3} \text{ mol}$$

CHECKPOINT

0.5 PTS

The total amount of electrons transferred by oxalic acid is $5.075 \times 10^{-3} \text{ mol}$

The total amount of electrons transferred by potassium permanganate is:

$$0.02030 \text{ mol/L} \times 21.84 \text{ mL} \times 5 = 2.217 \times 10^{-3} \text{ mol}$$

CHECKPOINT

0.5 PTS

The total amount of electrons transferred by potassium permanganate is: $2.217 \times 10^{-3} \text{ mol}$

So the number of electrons corresponding to 1.000 g of **X** is:

$$5.075 \times 10^{-3} \text{ mol} - 2.217 \times 10^{-3} \text{ mol} = 2.858 \times 10^{-3} \text{ mol}$$

Assume the chemical formula of **X** is Ag_xPbO_y , and assume that Ag and O are +1 and -2 valence, respectively, then:

$$(207.2 + 32.07 + 16.00 \times 4)/(107.9x + 207.2 + 16.00y) = 0.286/0.500 \text{ (a)}$$

$$1.000 \text{ g} \times (2y - x - 2)/(107.9x + 207.2 + 16.00y) = 2.858 \times 10^{-3} \text{ mol (b)}$$

CHECKPOINT

1 PTS

The two equations are $(207.2 + 32.07 + 16.00 \times 4)/(107.9x + 207.2 + 16.00y) = 0.286/0.500$ and $1.000 \text{ g} \times (2y - x - 2)/(107.9x + 207.2 + 16.00y) = 2.858 \times 10^{-3} \text{ mol}$

Solving equations (a) and (b) simultaneously yields:

$$x = 2.54 ; y = 3.03$$

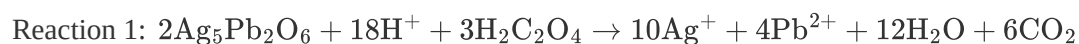
After converting to the simplest integer ratio, we can know that the chemical formula of **X** is: $\text{Ag}_5\text{Pb}_2\text{O}_6$

CHECKPOINT

1 PTS

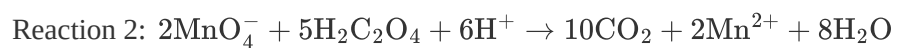
The chemical formula of **X** is: $\text{Ag}_5\text{Pb}_2\text{O}_6$

$\text{H}_2\text{C}_2\text{O}_4$ is oxidized to CO_2 , and the reaction equation is:


CHECKPOINT

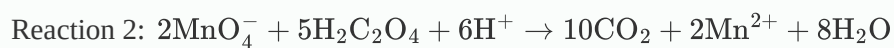
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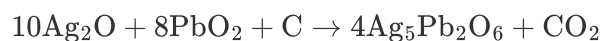


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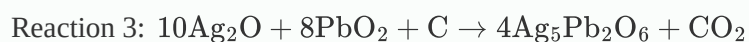


In reaction 3, Pb needs to be reduced, so activated carbon is oxidized to CO_2 :



CHECKPOINT

1 PTS



a=55, b=33, c=24, (a+b)/c=3.67, select C